RF Amplifier Verification Report

Automated RF Test and Verification System

Report generated: 2026-11-04 14:17:40 EST
Measurement source: Educational simulation; no physical RF instrument was connected.
Test mode: Simulated fault injected

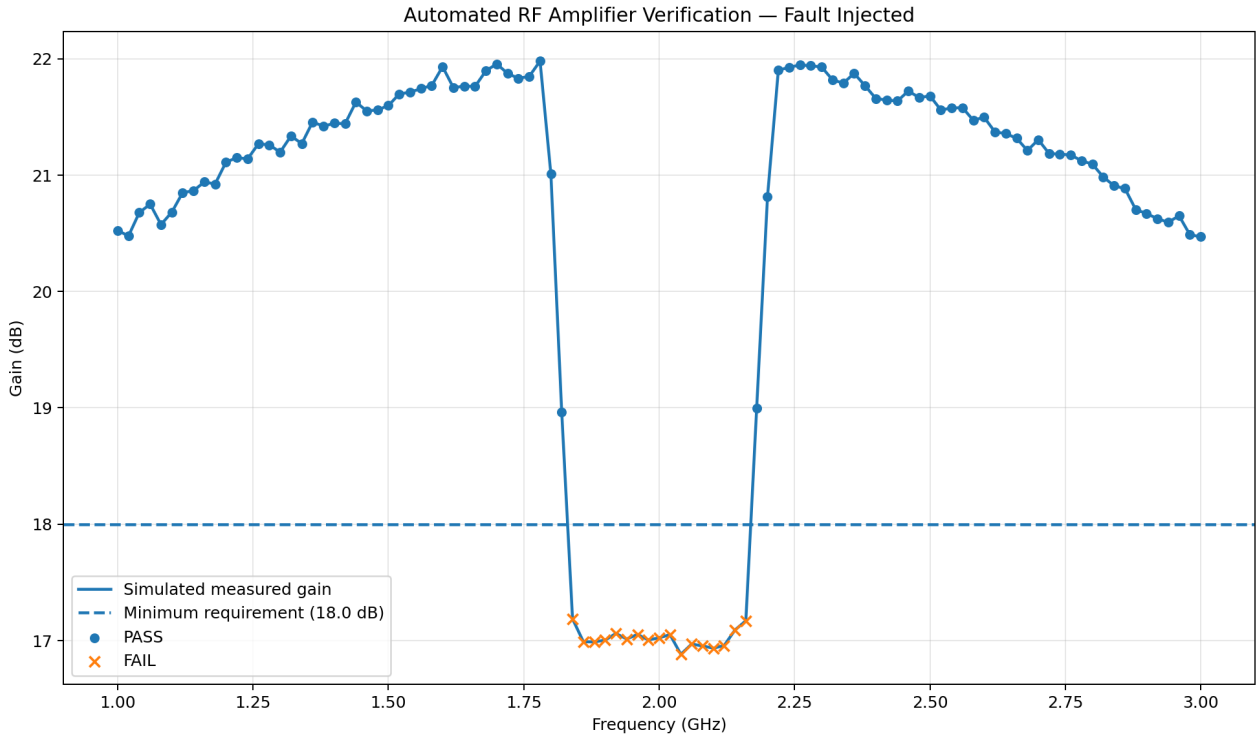
1. Test Configuration

Parameter	Value
Start frequency	1.00 GHz
Stop frequency	3.00 GHz
Frequency points	101
Minimum acceptable gain	18.00 dB
Injected fault drop	5.0 dB

2. Verification Summary

Metric	Result
Total measurement points	101
Passed measurements	84
Failed measurements	17
Minimum measured gain	16.881 dB
Worst-case margin	-1.119 dB
Overall result	**FAIL**

3. Frequency Response



The dashed line represents the configured minimum gain requirement. Measurements below this line fail the verification criterion.

4. Failed Measurement Points

Frequency (GHz)	Measured gain (dB)	Margin (dB)
1.840	17.185	-0.815
1.860	16.989	-1.011
1.880	16.988	-1.012
1.900	17.002	-0.998
1.920	17.060	-0.940
1.940	17.012	-0.988
1.960	17.052	-0.948
1.980	17.005	-0.995
2.000	17.023	-0.977
2.020	17.050	-0.950
2.040	16.881	-1.119
2.060	16.969	-1.031
2.080	16.953	-1.047
2.100	16.934	-1.066
2.120	16.956	-1.044
2.140	17.090	-0.910

Frequency (GHz)	Measured gain (dB)	Margin (dB)
2.160	17.169	-0.831

Engineering note: This report documents a software simulation. The amplifier response, injected fault, and acceptance threshold are illustrative and are not validated against actual satellite hardware specifications.